UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

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| COURSE | PRE CALCULUS |
| COURSE CODE | MAT133 |
| EXAMINATION | OCTOBER 2026 |
| TIME | 2 HOURS |

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

*This examination paper consists of 7 printed pages*

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## QUESTION 1

**a)** Solve the inequality 5x + 3 < -1. Express the answer in interval notation.

*(4 marks)*

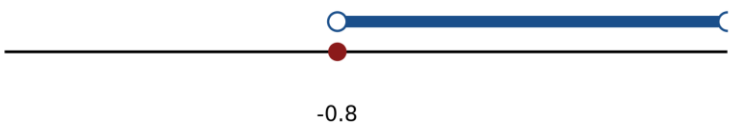



*Figure: solution on a number line.*

**b)** Solve the inequality |2x + 2| < 2. Express the answer in interval notation and on a number line.

*(5 marks)*



*Figure: solution on a number line.*

**c)** Sketch y = 2|x - 4| - 1. State the vertex, the y-intercept and any x-intercepts.

*(5 marks)*

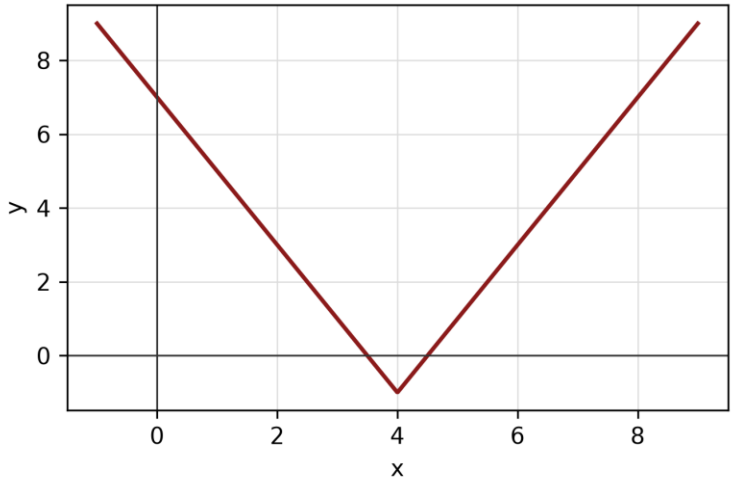



*Figure: absolute-value graph.*

## QUESTION 2

**a)** Determine algebraically whether f(x) = x^5 - 2x^3 - 5x is even, odd or neither. State the symmetry of its graph.

*(4 marks)*

**b)** Apply graph-transformation techniques to sketch $y = -(x - 2)^2 + 5$. State the vertex and the y-intercept.

*(5 marks)*

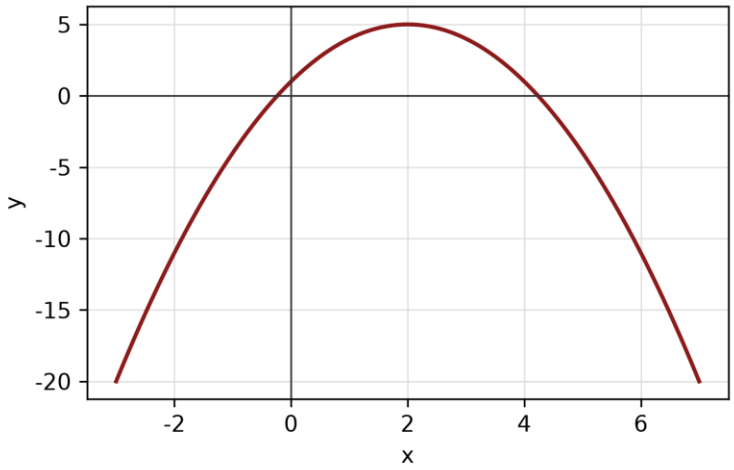



*Figure: transformed graph.*

**c)** Write the equation of the circle with centre (5, -4) and radius 7. Hence write its general equation and sketch the circle.

*(5 marks)*

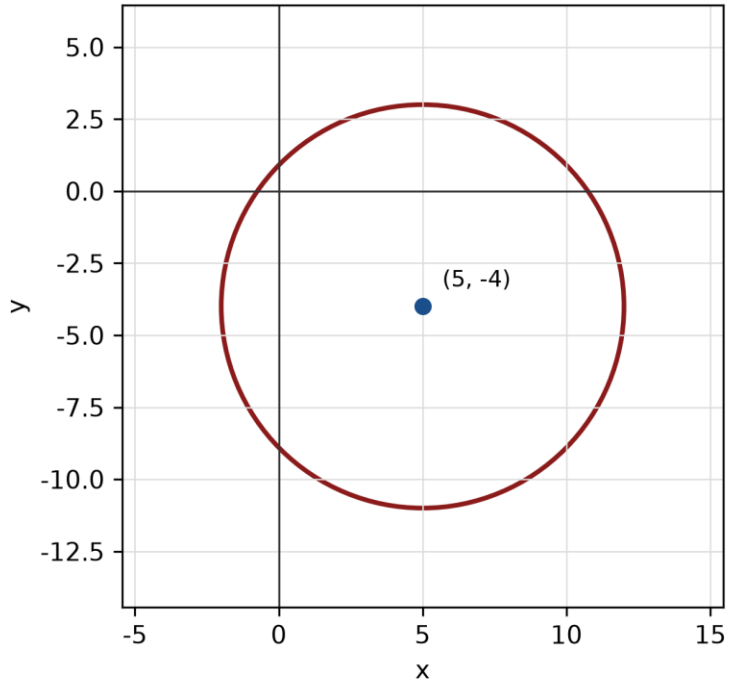



*Figure: circle.*

## QUESTION 3

**a)** Given $z_1 = -1 - i$ and $z_2 = -1 - 4i$, express $z_1/z_2$ in the form $a + bi$.

*(5 marks)*

**b)** Find the modulus and argument of $z = -3 + 4i$. Give the argument in degrees correct to one decimal place.

*(5 marks)*

**c)** Use synthetic division to divide $f(x) = -2x^3 + 5x^2 + 5x + 4$ by $x + 4$. State the quotient and remainder.

*(5 marks)*

## QUESTION 4

**a)** Solve the simultaneous equations
-3x - 2y = -10
-x - 2y = -10.

*(4 marks)*

**b)** Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (4, 0) and (0, 4).

*(5 marks)*

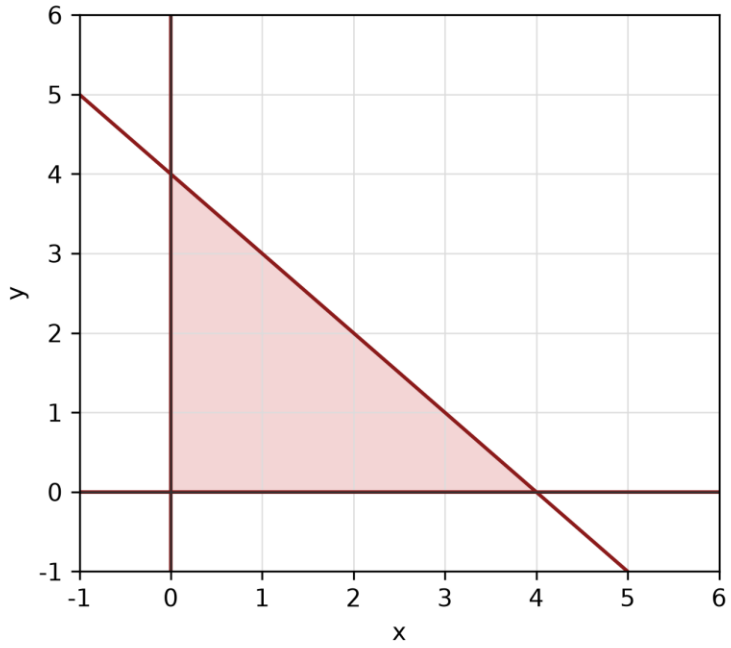



*Figure: shaded triangular region.*

**c)** Find the equation of the straight line passing through (-1, -1) and (-3, 3).

*(4 marks)*

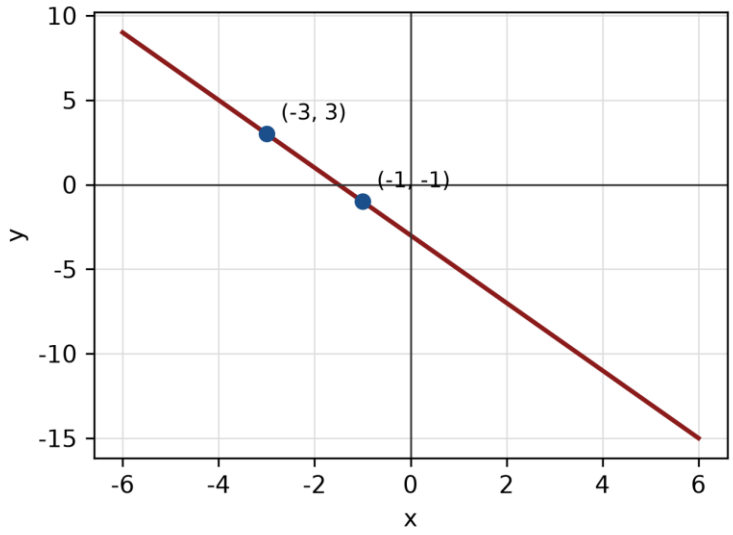



*Figure: line through the two points.*

## QUESTION 5

**a)** Solve the trigonometric equation $-\cos x - \cos(2x) - 1 = 0$ for $0° \leq x < 360°$.

*(5 marks)*

**b)** Find the perimeter of the sector with radius 6 cm and central angle 45°.

*(4 marks)*

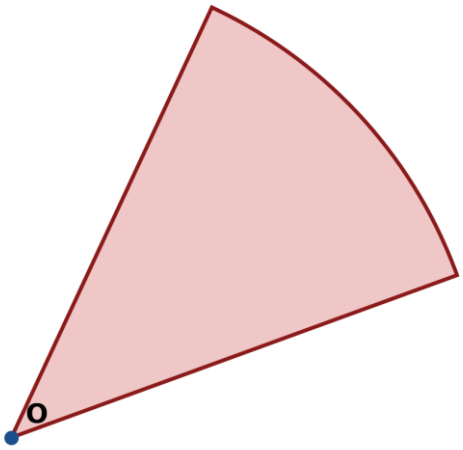



*Figure: sector.*

**c)** Solve the triangle with sides a = 15 cm, b = 36 cm and c = 39 cm. Find all three angles.
*(5 marks)*

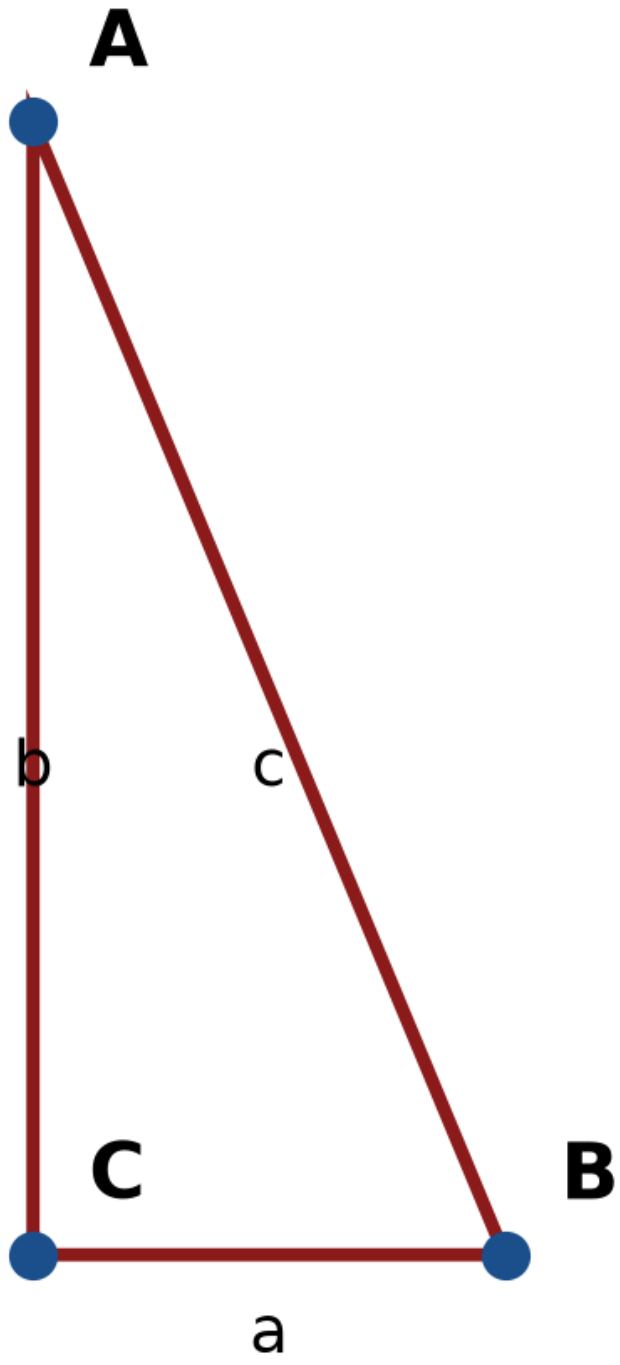



*Figure: right triangle.*

*END OF QUESTION PAPER*

# APPENDIX 1

*Formula sheet (as supplied with the paper).*

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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